

Homework 6

Release Date: 1404/9/16

Deadline: 1404/9/27

1 Sampling at the Nyquist Rate and Signal Reconstruction (20 pts)

Assume that the baseband signal $x(t)$ with bandwidth B is sampled at the Nyquist rate, producing the signal

$$x_1(t) = \sum_{k=-\infty}^{\infty} (-1)^k x(kT_s) \delta(t - kT_s).$$

1. Find the Fourier transform of $x_1(t)$ in terms of $X(f)$, and specify whether $x(t)$ can be reconstructed from $x_1(t)$. If not, explain how it can be reconstructed.
2. Assume that the signal $x(t) = 2B \text{sinc}(2Bt)$ is sampled using a periodic pulse train $p(t)$ with pulse width $\frac{1}{4B}$ and amplitude $4B$, and the sampling rate is greater than the Nyquist rate. Find the Fourier transform of the sampled signal and sketch it. Determine the frequency where the maximum attenuation occurs, and compute its value.

2 Lowpass Signal Sampled at the Nyquist Rate (15 pts)

A lowpass signal $x(t)$ with bandwidth of 50 Hz is sampled at the Nyquist rate and the resulting sampled values are

$$x(nT_s) = \begin{cases} -1, & -4 \leq n < 0, \\ 1, & 0 < n \leq 4, \\ 0, & \text{otherwise.} \end{cases}$$

1. Find $x(0.005)$.
2. Is this signal power-type or energy-type? Find its power or energy content.

3 Zero-Order Hold Circuit (20 pts)

A zero-order hold circuit in Fig. 1 is often used to reconstruct a signal $g(t)$ from its samples.

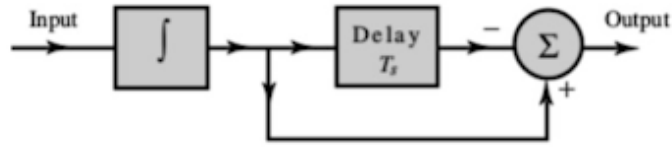


Figure 1: Zero-order hold reconstruction circuit.

1. Find the unit impulse response of this circuit.
2. Find the transfer function $H(\omega)$ and sketch $|H(\omega)|$.
3. Show that when a sampled signal $\bar{g}(t)$ is applied at the input of this circuit, the output is a staircase approximation of $g(t)$. The sampling interval is T_s .

4 PCM Coding of Compact Disc Audio (15 pts)

A compact disc (CD) records audio signals digitally by using PCM. Assume the audio signal bandwidth to be 15 kHz.

1. What is the Nyquist rate?
2. If the Nyquist samples are quantized into $L = 65,536$ levels and then binary coded, determine the number of binary digits required to encode a sample.
3. Determine the number of binary digits per second (bit/s) required to encode the audio signal.
4. For practical reasons discussed in the text, signals are sampled at a rate well above the Nyquist rate. Practical CDs use 44,100 samples per second. If $L = 65,536$, determine the number of bits per second required to encode the signal, and the minimum bandwidth required to transmit the encoded signal.

5 Quantization and SQNR in PCM (20 pts)

A signal can be modeled as a real, baseband, stationary random process $X(t)$ whose probability density function at any time t is

$$f_X(x) = \begin{cases} \frac{1}{3}, & -2 \leq x \leq 0, \\ \frac{1}{6}, & 0 < x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

The signal has a bandwidth of 5 kHz and we wish to transmit it using PCM over a channel with a guard band of 2 kHz. Assume that a uniform quantizer with 128 levels is used.

1. Find the bit rate (in bits per second) required to transmit the PCM signal.
2. Determine the minimum transmission bandwidth required for the bit stream.
3. Compute the output SQNR (Signal-to-Quantization-Noise Ratio) in dB.
4. If the channel transmission bandwidth is 70 kHz, what is the maximum SQNR that can be obtained at the receiver output?

6 FDM System Design (10 pts)

In some telephone systems that use frequency-division multiplexing (FDM), different signals require different FM modulators, and in some cases the message signals have different bandwidths. To simplify this situation, a special method is used in which the bandwidth of each message signal is adjusted so that it is proportional to the center frequency of its subcarrier. In this system, all FM subcarrier signals have the same deviation ratio, but for the i th subcarrier the message bandwidth W_i and the subcarrier center frequency f_i satisfy

$$f_i = \frac{W_i}{\alpha},$$

where α is a constant.

1. Show that the bandwidth of the FM subcarrier signal, denoted by B_i , is proportional to f_i . Also, derive a relation for f_{i+1} in terms of f_i if a guard band of width B_g is placed between adjacent channels.
2. Using the relation obtained in part (1), compute the subcarrier center frequencies when the initial values are

$$f_1 = 2 \text{ kHz}, \quad B_1 = 800 \text{ Hz}, \quad B_g \approx 400 \text{ Hz}.$$

Submission

Please submit your answers in a .pdf file named CS_HW6_studentID.pdf, replacing studentID with your actual student ID.